

Interdependencies among Asian Bond Markets: Impact of global and regional factors

**S.Sowmya, Doctoral Scholar, Department of Management Studies, IIT Madras and
Dr. P. Krishna Prasanna, Associate Professor, Department of Management Studies, IIT Madras**

Abstract

The paper investigates the global and regional influence on the sovereign bond yields of nine Asian economies. The term structure of interest rates is estimated by Dynamic Nelson Siegel model and three latent factors of the model (Viz. level representing long term interest rates, Slope representing short term interest rates and Curvature representing medium term interest rates) are extracted using Kalman filter. Global and regional factors are extracted using dynamic factor model in state space framework. The results show that long term interest rates in Japan, South Korea and Hong Kong are influenced by the global and regional factors. It is found that short term rates of Singapore and Hong Kong alone are influenced by US short term rates in the post crisis period. These results in general indicate that the financial integration doesn't reduce the control of monetary authorities on the front end of yield curve. The long term interest rates provide potential transmission channel to spread the crisis and financial contagion.

JEL Classification: E43, G12, G15, G01, C58

Key Words Yield curve, transmission channel, term structure model, Linkages

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1. Introduction

The globalization of financial markets and capital flows increased the interdependencies of the domestic term structure of interest rates (Holmes et.al. 2011). Earlier research studies on term structure of interest rates investigated the factors driving the yield curve in single country context (Litterman and Scheinkman, 1991; Diebold and Li, 2006; Alper et al., 2007; Koopman et al., 2010) and they found that three common factors such as level (representing long term interest rates), slope (representing short term interest rates) and curvature (representing medium term interest rates) explains the maximum variation in the bond yields. Bond markets quickly respond to the macroeconomic news; and term structure movements provide information about the future state of the economy (Wu, 2006). This has led to investigation of the influence of macroeconomic factors on the term structure of interest rates (Matsumura and Moreira, 2011; Lange, 2013; Kaya, 2013; Diebold et al., 2006, Wu, 2006). These studies noted significant influence of macroeconomic factors on future movements of term structure of interest rates.

Yield curve fluctuations across the countries are correlated (Jotikasthira et al. 2015). Davies (2007) found the evidence of common bond market trend among the developed economies. Diebold et al. (2008) constructed a global yield curve using Dynamic Nelson Siegel model (DNS) and found significant influence of global factors on the domestic yield curve. Byrne et al. (2012) also noted the influence of global factors especially in the long end of the yield curve. The growing evidence of deeper financial integration of Asian economies into global debt market has made Asian economies more sensitive towards the global and regional movements of the bond market (Turner, 2014). This phenomenon necessitates in exploring the impact of global and regional movements on the domestic yield curve among the Asian countries.

The Asian Financial crisis 1997-1998 indicated the importance of bond market in the region, suggested to reduce dependence on foreign borrowings and to channelize the regional savings. Since then a steady progress is achieved in the development of the Asian bond markets. Asian Bond Market Initiative (ABMI) and Asian Bond Fund (ABF) project supported the development of these bond markets. This development along with the dynamic economic growth in emerging Asian countries attracted large number of domestic and international investors in

recent years. The capital inflows in Emerging Asian government debt market have increased steadily since the global financial crisis (Pradhan et al.2011). Chandra and Unsal(2014) found that long term interest rates in Asia are driven by the global factors especially during the period of large capital inflows. As a follow up study, this paper investigates the vulnerability of the Asian countries yield curve to the global and regional movements. The current study contributes to the literature by examining the global and regional influence in the Asian countries term structure. The study has important implications to the monetary authorities in analyzing the efficacy of monetary policy and global investors for portfolio diversification

The study aims to identify the global and regional influence on the nine Asian countries term structure such as Japan, Hong Kong, Singapore, Korea, Malaysia, India, China, Philippines and Indonesia. The zero coupon yields of various maturity horizons are considered at monthly frequency from January 2003 to December 2013. The individual countries term structure is modeled using Dynamic Nelson Siegel model as in Diebold et al. (2006). Dynamic Nelson Siegel model decomposes the term structure in to three latent factors representing long term interest rates (level), short term interest rates (slope) and medium term interest rates (curvature). These latent factors are estimated from the model using Kalman filter. The global and regional factors are extracted using dynamic factor model in state space framework. The global factor is constructed by extracting the common factor from the developed countries such as US, UK, Germany and Australia and the regional factor using Japan, Singapore, Malaysia and Hong Kong. The influence of global and regional factor on the domestic term structure is estimated using regression analysis. Further the sample is divided into two phase's viz., pre crisis period and post crisis period to analyze influence and evolution of Asian yield curve before and after the global financial crisis, 2008.

The results of the study indicate both global and regional factors influences the long term interest rates in Japan, Hong Kong, Korea and Indonesia. Long term interest rates of Singapore and Philippines are influenced by regional factor than the global factor. However the long term rates of Malaysia, China and India diverges with the global and regional movements. On the other hand global and regional factors don't influence the short term interest rates of any of the Asian countries during the sample period.

Short term interest rates of Singapore and Hong Kong are influenced by the US short term interest rates especially after the Global financial crisis, 2008. The result indicates that the financial integration does not reduce the control of monetary authorities in short term interest rates among Asian economies. The long term interest rates provide the potential channel for international monetary and financial transmission. It offers diversification opportunities to the global investors in investing in Asian bond markets. The results are robust to changes in the variations of the model.

The contribution of the study is in three fold. It is first to examine the global and regional influences on the term structure of interest rates of emerging Asian economies. This relationship helps in understanding the co movement of other financial assets. Earlier studies (Diebold et al. 2008; Abritti et al. 2013) investigated the global influences in the developed countries term structure ignoring regional influences. Bae and Kim (2011) analyzed the global and regional factors upon Japan, South Korea, Hong Kong and Singapore. The current study extends the work of Bae and Kim (2011) by examining the regional and global influences in the emerging Asian economies and uses the array of zero coupon yields to model the term structure of interest rates for the sample countries. Hence the results of the study are more robust in extracting the latent factors compared to the previous work.

Secondly the study helps in understanding the vulnerability of Asian bond markets to the movements in the global and regional bond yields. During the global financial crisis 2008, the low term premium in the advanced economies and stronger economic fundamentals in Emerging economies has attracted the global investors to diversify their portfolio in the emerging markets assets. This led to increase in the capital flows on emerging economies assets (Pradhan et al. 2011). Thus understanding the movement of Asian bond markets to the movements of global and regional bond markets helps the global investors to plan diversification strategies and to exercise the choice of asset allocation during the times of high market volatility.

Thirdly the study contributes the literature on identifying the channels through which the crises spreads. The transmission channels are in general divided into two groups. The first channel could be trade related fundamental economic factors (Berkmen et al.2012) and second channel could be Investor driven stock market linkages (Xu and Hamori, 2012; Samarakoon,2011). While Brekmen et al.(2012) provides evidence of trade related transmission channel,. Xu and

Hamori(2012) found that stock market linkages as the transmission channel. The present study identifies apart from stock market, yield curve linkages in the bond market also provides another channel to spread the financial contagion.

The remainder of the paper is organized as follows, Section 2 bond market integration, Section 3 presents the model, Section 4 describes the data, Section 5 presents the empirical results Section 6 summarizes and concludes.

2. Bond Market Integration

The previous studies in the bond market integration focused mainly on the co movement dynamics of bond returns. Piljak (2013) analyzed the co movement dynamics between ten emerging and four frontier markets with the US bond market. Their study found that domestic monetary policy and domestic macro-economic factors are more influential than the global factors. Bunda et al.(2009) analyzed the co movement among the emerging market bond returns and found that emerging market return are driven more by external factors during the period of crisis. Yang and Hamori (2014) examined interdependence of Poland, Czech Republic and Hungary bond markets with Germany. Their study found that integration is more in long term interest rates than the short term interest rates.

The studies related to financial integration in Asian bond market is thin. Vo (2009) investigates the integration in Asian bond market during the Asian Financial crisis period and found that there is low degree of financial integration between Australian/US bond yield and the Asian Bond market.

Another strand of literature constructed the global yield curve and explored its impact on the domestic yield curve. In particular Diebold et al. (2008) constructed global yield curve using hierarchical dynamic factor model using government yields of US, UK, Germany and Japan. The variations in the country's bond yields are decomposed into global factors and the country specific factors. Their study found that the Global level and slope factors contribute to the significant variation in the yield rates in the domestic market. Morita and Bueno (2008) examined the influence of global factors on the emerging markets (South Africa, South Korea and Mexico) and found emerging market yields are well connected with the global factors. Bae and Kim (2011) investigated the influence of global and regional factors among the East Asian

countries. Their results highlighted that global factor exhibits significant contribution in comparison with the regional factors. Jaramillo and Weber (2013) studied the determinants of domestic bond yields in emerging markets. They used principal component analysis to extract the common factors of the domestic bond yields. Their result highlighted the existence of two common factors. Abritti et al. (2013) used factor augment vector auto regression to extract the global factors from the yield curve of seven open economies (i.e. Canada, United Kingdom, Japan, Germany, Australia, New Zealand and Switzerland) and found that global factors are ultimate drivers of the domestic yield curve and term premia.

3. Methodology

3.1. Dynamic Nelson Siegel Model

The Nelson and Siegel (1987) model is used to estimate the cross sectional yields at any point of time. The yield curve is represented as

$$y(\tau) = \beta_1 + \beta_2 \left(\frac{1-e^{-\lambda\tau}}{\lambda\tau} \right) + \beta_3 \left(\frac{1-e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) \quad (1)$$

Where $y(\tau)$ is the yield and τ denotes maturity of the bond. $\beta_1, \beta_2, \beta_3$ and λ are parameters. Diebold and Li(2006) represented the nelson Siegel model in dynamic form as

$$y(\tau) = L_t + S_t \left(\frac{1-e^{-\lambda\tau}}{\lambda\tau} \right) + C_t \left(\frac{1-e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) \quad (2)$$

Where, L_t, S_t and C_t are the latent factors of the yield curve. L_t is the level factor of the yield curve which represents the long term yields. S_t is the slope factor of the yield curve representing the short term yields. C_t is the curvature factor of the yield curve representing the medium term yields. The three latent factors are extracted using kalman filter in state space form for each country. The state space form contains two equations. The measurement equation relates the bond yields and the latent factors and written as:

$$\begin{pmatrix} y_t(\tau_1) \\ y_t(\tau_2) \\ \vdots \\ y_t(\tau_N) \end{pmatrix} = \begin{pmatrix} 1 & \left(\frac{1-e^{-\tau_1\lambda}}{\tau_1\lambda}\right) & \left(\frac{1-e^{-\tau_1\lambda}}{\tau_1\lambda} - e^{-\tau_1\lambda}\right) \\ 1 & \left(\frac{1-e^{-\tau_2\lambda}}{\tau_2\lambda}\right) & \left(\frac{1-e^{-\tau_2\lambda}}{\tau_2\lambda} - e^{-\tau_2\lambda}\right) \\ \vdots & \vdots & \vdots \\ 1 & \left(\frac{1-e^{-\tau_N\lambda}}{\tau_N\lambda}\right) & \left(\frac{1-e^{-\tau_N\lambda}}{\tau_N\lambda} - e^{-\tau_N\lambda}\right) \end{pmatrix} \begin{pmatrix} L_t \\ S_t \\ C_t \end{pmatrix} + \begin{pmatrix} \varepsilon_t(\tau_1) \\ \varepsilon_t(\tau_2) \\ \vdots \\ \varepsilon_t(\tau_N) \end{pmatrix} \quad (3)$$

The transition equation relates the dynamics of the latent factor and written as:

$$\begin{pmatrix} L_t - \mu_l \\ S_t - \mu_s \\ C_t - \mu_c \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} L_{t-1} - \mu_l \\ S_{t-1} - \mu_s \\ C_{t-1} - \mu_c \end{pmatrix} + \begin{pmatrix} \eta_t(L) \\ \eta_t(S) \\ \eta_t(C) \end{pmatrix} \quad (4)$$

Where L_t , S_t and C_t are the latent factors in the measurement equation which are extracted using Kalman filter.

3.2. Extraction of Global and Regional Factors using dynamic factor model

Stock and Watson (1991) developed a dynamic factor probability model that captures the co movement of many macro economic variables into a single unobserved factor using the application of state space model and Kalman filter.

The global yield rates are not observable in reality; hence the dynamic factor model is used to capture the co movement of developed countries bond yields into a single global factor. This involves two steps. In first step the country level latent factors of dynamic Nelson Siegel model is extracted using Kalman filter. In Second step the commonality of each latent factor is captured using dynamic factor model and termed as global level and global slope factor respectively. Similarly for regional factor, the co-movement of Asian countries level and slope factor is used to extract the regional level and regional slope factors.

The level, slope factor of yield curve are considered to construct the global level and global slope factors. The curvature factor is ignored as it lacks the clear link with macro economic variables and there is less correlation among the curvature factor across the countries (Diebold et. al., 2006)

The country's latent factors loads into the global and regional latent factors as

$$l_{it} = \alpha_i^l + \beta_i^l l_{gt} + \beta_i^l l_{rt} + \varepsilon_{it}^l \quad (5a)$$

$$s_{it} = \alpha_i^s + \beta_i^s s_{gt} + \beta_i^s s_{rt} + \varepsilon_{it}^s \quad (5b)$$

Where (α_i^l, α_i^s) are constant, $(\beta_i^l l_{gt}, \beta_i^s s_{gt})$ are the countries loading on the global latent factors and $(\beta_i^l l_{rt}, \beta_i^s s_{rt})$ are the countries loading on the regional factors and $(\varepsilon_{it}^l, \varepsilon_{it}^s)$ represents domestic idiosyncratic factors.

The country's idiosyncratic factors follow first order autoregressive dynamics for both level and slope factors

$$\begin{pmatrix} \varepsilon_{it}^l \\ \varepsilon_{it}^s \end{pmatrix} = \begin{bmatrix} \varphi_{i,11} & \varphi_{i,12} \\ \varphi_{i,21} & \varphi_{i,22} \end{bmatrix} \begin{pmatrix} \varepsilon_{i,t-1}^l \\ \varepsilon_{i,t-1}^s \end{pmatrix} + \begin{pmatrix} u_{it}^l \\ u_{it}^s \end{pmatrix} \quad (6)$$

Where u_{it}^l and u_{it}^s are the respective disturbances. Therefore the linear regression of Eqn. 5a and 5b is modeled with AR (1) errors.

4. Data Description

The monthly yields of maturities 3,6 12,24,36,48,60,72, 84,96,108,120 months for US, UK, Germany, Japan, Australia, Singapore, Malaysia, Hong Kong, Korea, India, Indonesia, China and Philippines are considered for the period January 2003 to December 2013 at monthly frequency. The standard bootstrapped zero coupon yields are taken from the Bloomberg database except for India. Indian yields are bootstrapped based on the government securities trading data provided by the National Stock Exchange.

Table1: Descriptive Statistics

Maturity	Mean	Std. Dev	Minimum	Maximum	$\rho(1)$	$\rho(12)$
Japan						
3months	0.19	0.21	0.002	0.69	0.98	0.55
1year	0.23	0.23	0.01	0.83	0.98	0.58
5year	0.67	0.37	0.12	1.53	0.96	0.59
10 year	1.32	0.36	0.57	1.97	0.94	0.47
HongKong						
3months	1.12	1.39	0.03	4.09	0.97	0.57
1year	1.34	1.45	0.03	4.43	0.98	0.64
5year	2.46	1.33	0.24	4.80	0.97	0.72
10 year	3.13	1.27	0.61	5.01	0.97	0.64

Singapore						
3months	1.04	0.93	0.12	3.28	0.98	0.69
1year	1.10	0.93	0.15	3.21	0.99	0.71
5year	1.82	0.88	0.46	3.45	0.97	0.71
10 year	2.72	0.67	1.40	4.21	0.95	0.43
Malaysia						
3months	2.84	0.45	1.78	3.51	0.95	0.15
1year	2.89	0.48	1.77	3.82	0.96	0.13
5year	3.68	0.38	2.91	4.63	0.90	-0.09
10 year	4.14	0.52	3.17	5.41	0.91	0.07
Korea						
3 months	3.86	1.55	1.90	17.58	0.43	0.13
1 year	3.96	0.95	2.54	7.11	0.92	0.40
5 year	4.52	0.84	2.68	6.16	0.94	0.46
10 year	4.82	0.84	2.90	6.15	0.94	0.48
India						
3months	6.2	1.72	3.07	11.08	0.95	0.30
1year	6.37	1.49	3.57	9.49	0.94	0.31
5year	6.84	1.10	3.83	8.85	0.88	0.41
10 year	7.27	0.93	4.89	8.91	0.88	0.36
China						
3months	2.42	0.83	0.96	4.96	0.87	-0.01
1year	2.50	0.79	0.99	4.25	0.94	-0.06
5year	3.33	0.67	1.89	4.79	0.92	-0.22
10 year	3.83	0.63	2.68	5.45	0.92	-0.20
Indonesia						
3months	7.94	2.60	3.89	15.20	0.95	0.23
1year	8.14	2.63	3.89	15.33	0.96	0.28
5year	9.73	2.85	4.75	16.85	0.96	0.46
10 year	10.40	2.78	5.37	17.99	0.95	0.48
Philippines						
3months	4.40	2.33	0.12	8.15	0.96	0.64
1year	5.36	2.71	0.60	9.97	0.97	0.70
5year	7.46	2.94	2.43	13.05	0.97	0.66
10 year	8.62	3.10	2.95	14.49	0.97	0.62

Notes: This table provides the descriptive statistics of monthly bond yields at representative maturities for the period January 2003 to December 2013. ρ represents the autocorrelation at displacement t .

Table 1 presents the descriptive statistics of monthly bond yields of representative maturities of the Asian markets. The yield curve was upward sloping for all the Asian markets. The volatility is higher for long term maturities. The average yield is high in Indonesia ranging from 7.94% to 12.01% and low in Japan ranging from 0.19% to 2.30%. There is high persistence among the yields in the Asian markets.

The yield curve for each country is modeled by Dynamic Nelson Siegel model (DNS) as in Diebold et al. (2006). The DNS model quantifies the yield curve into three latent factors viz., level, slope and curvature which are extracted using Kalman filter in state space form. Table 2 represents the descriptive statistics of the extracted level, slope and curvature factors for each country.

Table 2: Descriptive statistics of extracted level and slope factors for Asian countries

	Mean	Std. Dev	Minimum	Maximum	Jarque-Bera t- statistics	P values	$\rho(1)$	ADF t- statistics	P values
Japan									
Level	3.20	0.70	1.72	5.22	2.97	0.23	0.93	-1.68	0.44
Slope	-3.01	0.79	-5.22	-1.69	7.12	0.03**	0.95	-1.52	0.52
Curvature	-2.94	1.27	-5.15	-0.09	7.57	0.02**	0.96	-2.31	0.17
Hong Kong									
Level	3.93	1.41	0.87	6.73	1.49	0.47	0.96	-2.53	0.11
Slope	-2.88	1.72	-6.67	-0.38	7.76	0.02**	0.96	-1.92	0.32
Curvature	-2.13	2.11	-6.20	1.65	7.96	0.02**	0.93	-2.14	0.23
Singapore									
Level	4.35	0.93	2.97	6.72	10.42	0.01	0.92	-3.48475	0.00***
Slope	-3.27	1.51	-6.03	-0.01	5.34	0.07*	0.96	-2.13887	0.23
Curvature	-3.69	2.05	-7.66	0.07	5.80	0.06*	0.96	-1.35491	0.60
Malaysia									
Level	4.77	0.66	3.50	6.84	9.50	0.01***	0.90	-3.83255	0.00***
Slope	-1.94	1.06	-4.47	-0.15	10.92	0.00***	0.95	-2.59003	0.09**
Curvature	-1.78	0.90	-4.33	0.48	0.47	0.79	0.83	-2.33224	0.16
Korea									
Level	5.10	0.85	3.12	6.44	14.41	0.00***	0.94	-1.77	0.39
Slope	-1.28	1.31	-4.98	3.23	17.69	0.00***	0.85	-3.20	0.02**
Curvature	-0.98	1.14	-6.47	3.82	150.07	0.00***	0.59	-5.41	0.00***
India									
Level	7.69	0.63	6.04	8.79	18.55	0.00***	0.94	-2.73	0.07*
Slope	-1.54	1.68	-5.31	2.80	5.68	0.06*	0.96	-1.86	0.35
Curvature	-0.27	1.75	-9.23	3.90	120.21	0.00***	0.78	-3.33	0.01***
China									
Level	4.31	0.79	3.17	6.69	35.69	0.00***	0.93	-3.05	0.03**
Slope	-1.82	0.86	-3.60	0.08	4.44	0.11	0.92	-2.47	0.13
Curvature	-2.42	1.35	-6.02	-0.29	17.83	0.00***	0.92	-3.18	0.02**
Indonesia									
Level	11.86	3.04	6.26	20.17	1.57	0.454	0.91	-2.04	0.26
Slope	-19.52	6.78	-35.45	-10.5	10.33	0.00***	0.96	-2.18	0.21
Curvature	-11.08	4.40	-20.5	-4.75	12.31	0.00***	0.33	-2.01	0.27
Philippines									
Level	9.61	3.14	3.16	15.36	4.73	0.09*	0.96	-1.47	0.54
Slope	-5.34	1.63	-9.17	-2.04	6.26	0.04***	0.91	-3.05	0.03***
Curvature	0.304	3.82	-8.04	6.83	7.01	0.03**	0.92	-2.86	0.05**

Notes: This table provides the descriptive of the latent factors extracted from Dynamic Nelson Siegel model applying kalman filter. Augmented Dickey fuller(ADF) test is used to check the stationarity of the series. Jarque-Bera tests the normality of the series. Rejection of null hypothesis at 1%, 5% and 10% levels are denoted by ***, ** and * respectively. $\rho(t)$ denotes sample autocorrelation at displacement t.

The average level factor is highest in Indonesia followed by Philippines and lowest in Japan. The average slope factor is highest is Korea and lowest in Indonesia. The volatility of the factors is high in Hong Kong, Indonesia and Philippines. The curvature factor is not considered for the further analysis as it is found that it doesn't influence the macroeconomic fundamentals (Diebold et.al, 2006) and the correlation between the curvature factor across the economies is found to be less. Hence the level and slope factor are considered to analyze the influence of global and regional movements across the Asian economies.

Correlations of latent factors

Table 3: Correlation Matrix of Level factors across countries

	US	UK	Germany	Australia	Japan	Singapore	Hong Kong	Korea	Malaysia	India	China	Philippines	Indonesia
US	1												
UK	.739***	1											
Germany	.713***	.679***	1										
Australia	.803***	.688***	.697***	1									
Japan	.485***	.402***	.383***	.473***	1								
Singapore	.615***	.401***	.513***	.621***	.566***	1							
Hong kong	.617***	.474***	.482***	.591***	.339***	.621***	1						
Korea	.451***	.347***	.302***	.546***	.310***	.495***	.423***	1					
Malaysia	.309***	.122	.311***	.271***	.121	.479***	.292***	.269***	1				
India	.193**	.125	.190**	.223**	.086	.218**	.159	.221**	.356**	1			
China	.097	.103	.173	.102	-.095	.134	.214**	.185**	.215**	.110	1		
Philippines	.155	.184**	.114	.206**	.213**	.188**	.147	.166	.211**	.224**	-.057	1	
Indonesia	.172	.035	.082	.072	.035	.101	.054	.128	.098	.097	-.110	.203**	1

Table 3 presents pair wise correlation between the level factors across the countries. Singapore, Japan, Hongkong and Korea long term factor are highly correlated with all the long term factor of developed markets. It is also noted that Singapore, Japan, Hongkong, Korea, Malaysia and India rates are highly correlated with US and Australia

Table 4: Correlation matrix of slope factor across countries

	US	UK	Germany	Australia	Japan	Singapore	Hong Kong	Korea	Malaysia	India	China	Philippines	Indonesia
US	1												
UK	.180**	1											
Germany	.411***	.226*	1										
Australia	.185**	.492**	.282***	1									
Japan	.257***	.018	.317***	.093	1								
Singapore	.074	-.003	.127	.094	.204**	1							
Hong Kong	.198**	.101	-.021	.074	.108	.241***	1						
Korea	.169	-.046	.162	.008	.324***	.059	.098	1					

Malaysia	.392***	.056	.157	.072	.144	.339***	.196**	.141	1				
India	.060	.112	.137	.001	.131	.102	.026	.092	.062	1			
China	-.025	.003	.022	-.031	-.029	.101	.063	.067	.032	.097	1		
Philippines	-.044	.098	.110	.200***	-.019	.163	.057	.135	.184***	-.055	.168	1	
Indonesia	.126	.199** **	.122	.077	.035	.189***	.073	.035	.033	-.062	-.093	-.075	1

Table 4 presents the pair wise correlation among the slope factors. It is found that Hong Kong and Malaysia has significant correlation with US short term rates. While Singapore and Korea short term rates are correlated with Japan.

4.1. Preliminary analysis for extraction of Global factors

The preliminary assessment of existence of common global factor is found by conducting the principal component analysis. The level and slope factor of both developed and Asian countries are considered in the principal component analysis to check the existence of one dominant factor.

Table 5: Extraction of global level factor using Principal Component Analysis

Level Factor – Global				Slope Factor – Global			
Component	Eigenvalue	Variance Proportion	Cumulative Proportion	Component	Eigenvalue	Proportion	Cumulative
PC 1	7.150	0.550	0.55	PC 1	6.589	0.507	0.507
PC 2	1.795	0.138	0.688	PC 2	2.770	0.213	0.720
PC 3	1.561	0.120	0.808	PC 3	1.128	0.087	0.807
PC 4	0.809	0.062	0.870	PC 4	0.949	0.073	0.880
PC 5	0.519	0.040	0.910	PC 5	0.536	0.041	0.921
PC 6	0.435	0.033	0.944	PC 6	0.302	0.023	0.944
PC 7	0.239	0.018	0.962	PC 7	0.245	0.019	0.963
PC 8	0.162	0.013	0.975	PC 8	0.194	0.015	0.978
PC 9	0.108	0.008	0.983	PC 9	0.109	0.008	0.986
PC 10	0.086	0.007	0.989	PC 10	0.080	0.006	0.993
PC 11	0.065	0.005	0.994	PC 11	0.060	0.005	0.997
PC 12	0.043	0.003	0.998	PC 12	0.023	0.002	0.999
PC 13	0.030	0.002	1	PC 13	0.014	0.001	1.000

Notes: This table reports the results of the Principal component analysis for the sample of four developed (US, UK, Germany and Australia) and nine Asian economies to find the existence of one dominant global factor. The analysis is done separately for both the level and slope factors.

Table 5 presents the Principal component analysis of level and slope factor for all the countries. The first principal component for level factor explains 55% of variation which is dominant and indicates the existence of commonality in the level factor among the countries yield curve. Similarly the first principal component of the slope factor explains 51% of the slope variation. Hence there is a single common component exist both in the level and slope factors. The results are in line with the study made in developed markets by Diebold et al (2008) and found the existence of dominant factor both in level and slope. The US, Australia, Germany and UK explains the majority of the variation in the first principal component of level and slope factor. Thus these four countries are chosen to extract the global component for the level and slope factors using dynamic factor model.

4.2. Regional factors

The principal component analysis is estimated for the Asian countries to check the possible existence of one dominant component in the level and slope factors representing regional factor.

Table 6 : Extraction of Regional level factor using Principal Component Analysis

Regional Component – Level Factor				Regional Component – Slope factor			
Component	Eigen value	Variance Proportion	Cumulative Proportion	Component	Eigen value	Proportion	Cumulative
PC 1	4.740	0.5267	0.5267	PC 1	4.794	0.533	0.533
PC 2	1.473	0.1637	0.6903	PC 2	1.417	0.157	0.690
PC 3	1.019	0.1132	0.8035	PC 3	0.938	0.104	0.794
PC 4	0.636	0.0707	0.8742	PC 4	0.809	0.090	0.884
PC 5	0.492	0.0546	0.9288	PC 5	0.482	0.054	0.938
PC 6	0.277	0.0308	0.9595	PC 6	0.233	0.026	0.964
PC 7	0.218	0.0243	0.9838	PC 7	0.163	0.018	0.982
PC 8	0.100	0.0111	0.9949	PC 8	0.099	0.011	0.993
PC 9	0.046	0.0051	1	PC 9	0.066	0.007	1.000

Notes: This table reports the results of the Principal component analysis of nine Asian economies to find the existence of one dominant regional factor. The analysis is done separately for both the level and slope factor.

The first principal component of level factor explains a 52.7 % variation which indicates the existences of one dominant factor among Asian countries. Similarly the first principal component of slope factor for Asian countries exhibits 53.3% of variation. Thus both level and slope factor of Asian countries confirms one dominant regional factor.

4.3. Global and Regional factor extraction

The results of the principal component analysis indicate the existence of global and regional components. The global and regional factors are extracted applying dynamic factor model in state space framework using kalman filter. The level and slope factors of US, UK, Germany and Australia are considered to extract the global factor. Singapore, Hong Kong, Japan and Malaysia are considered to extract the regional factors as these countries contributed the maximum variation in the first component¹. The global and regional factor for both level and slope is extracted using dynamic factor model as in Stock and Watson (1991). Further the influence of global and regional factors on the countries latent factors are analyzed using regression model as in Eqn(5a) and (5b).

5. Empirical Results

5.1. Role of global and regional influence

5.1a: Association across long term interest rates

The level factor representing long term interest rate indicates the future economic conditions and is driven by investor preferences and risk attitudes (Yang, 2005). The long term interest rate represents fundamental financial stability in an economy (Turner, 2014). Long term interest rates fluctuate because of changes in term premium and the capital flows in to the economy. Thus it is expected that global factor influences the long term interest rates among the economies. The influence of global and regional factors on the level factor extracted for each Asian country is estimated using the Regression analysis with AR (1) errors.

Table 7: Estimates of Global and regional level factors

$Level_{countires,t} = \alpha + \beta_{1t} global^{level} + \beta_{2t} regional^{level} + \varepsilon_t^{level}$				$\varepsilon_{countires,t}^{level} = \varepsilon_{countires,t-1}^{level} + v_t^{level}$		
		Global factor	Regional factor	Error	Lag of error term	Residual
$L_{japan,t}$	-0.00 (0.006)	0.020*** (0.005)	0.023*** (0.004)	$\varepsilon_{japan,t}^1$	0.16* (0.08)	0.002 (0.01)
$L_{Singapore,t}$	4.50 (0.48)	0.012 (0.02)	0.077*** (0.02)	$\varepsilon_{Singapore,t}^1$	0.94*** (0.08)	0.15 (0.02)
$L_{Hong Kong,t}$	-0.005 (0.01)	0.028*** (0.00)	0.03*** (0.00)	$\varepsilon_{Hongkong,t}^1$	0.37*** (0.08)	0.04 (0.02)

¹ The regional factors are extracted using Singapore, Hongkong, Japan and Korea. Both the results yielded the same results.

$L_{Korea,t}$	-0.002 (0.004)	0.001*** (0.003)	0.01*** (0.002)	$\varepsilon^l_{Korea,t}$	0.029 (0.09)	0.183 (0.003)
$L_{Malaysia,t}$	4.86 (0.004)	-0.018 (0.017)	0.009 (0.02)	$\varepsilon^l_{Malaysia,t}$	0.90*** (0.04)	0.20 (0.02)
$L_{india,t}$	-0.00 (0.003)	0.002 (0.001)	0.001 (0.001)	$\varepsilon^l_{india,t}$	0.46*** (0.07)	-0.30 (0.001)
$L_{china,t}$	4.07 (0.25)	-0.007 (0.01)	0.015 (0.01)	$\varepsilon^l_{china,t}$	0.93*** (0.02)	0.10 (0.01)
$L_{philippines,t}$	- 0.009 (0.008)	0.005 (0.003)	0.006** (0.003)	$\varepsilon^l_{philippines,t}$	0.50*** (0.07)	- 0.03 (0.01)
$L_{indonesia,t}$	- 0.002 (0.004)	0.008* (0.005)	0.005* (0.003)	$\varepsilon^l_{indonesia,t}$	- 0.61*** (0.07)	0.04 (0.01)

Notes: The table reports the estimates of global and regional influence on country's level factor. l_{gt} represents the global level factor, l_{rt} represents the regional level factor. ε^l represents the error term. v^l is defined as u_i/σ_i . The figure in the parenthesis represents the standard errors. *, ** and *** shows significance at 10%, 5% and 1 % respectively. The linear regression with AR (1) errors are estimated using Cochrane Orcutt procedure and MLE. Both yielded the same results.

The influence of global level factor is positive for all the countries except for china and Malaysia. The influence of regional level factor is positive across all the Asian countries. However, the degree of influence of global and regional factors differs across the Asian countries. Japan level factor has positive and significant influence on the global and regional level factor. The country specific level factor loading for Japan is relatively lower which indicates Japan level factor is less influenced by country specific factors. Japanese bond market is one among the most developed and liquid bond markets in the world and long term interest rates are more influenced by the external factors than the domestic factors.

Korea and Hong Kong also exhibits positive and significant influence both on the global level and regional level factors. Hong Kong and Korean bond markets are open and there are no restrictions for the foreign investors investing in these debt markets. This result provides evidence to the premise that movements in the long term rates are driven by global investor's cash flows in these countries.

On the other hand the long term interest rates among India, Malaysia and China exhibit independent behavior. Neither global nor regional factors have significant influence upon these markets. Chinese market is not influenced as their bond market still operates under tight controls. This result is consistent with the study of Glick and Hutchison (2013). Their study found that the linkages between Asian economies and long term interest rates of china are weak. Similar study by Chueng et al (2008) found that US interest rates doesn't drives the

Chinese interest rates. The Chinese interest rates are found to be independent with the rest of the world economies. Similarly India is also not influenced by the external factors as their debt markets are not as open as other Asian economies. Indian bond market also operates in a regulated environment with Government's intervention. Malaysia is not influenced by the global or the regional factors despite being an open economy. The country attracts high level of investments from foreign investors in their local currency bonds (32.6% at the end of March 2013). Despite being the hot spot among foreign investors for investment, the economy is not influenced by the external factors. This is due to the fact that Malaysian bonds are considered safe as they maintain moderate level of inflation and capital account surplus. The selloff of local currency bonds in Malaysia by the foreign investors was noted to be much lower during the recent global turmoil (Source: Policy Brief, 2013, Asian Development Bank).

Singapore and Philippines are influenced by regional than the global factors. Singapore follows macro prudential policy measures to keep them away from the systemic risk caused by global turmoil.

Among the emerging economies, Indonesia is found to have positive and significant relationship with both global and regional factors. This is because there is increased foreign ownership in the Indonesian sovereign bonds. The foreign ownership of the Indonesian sovereign bonds accounts for 38.6% in March 2015 (Source: Bank Indonesia). The Indonesian bonds are considered riskier due to the rising inflation and current account deficit. There was huge sell off by the foreign investors during the periods of global turmoil (Source: Policy Brief, Asian Development Bank)

5.2. Sub sample analysis

The stability of the analysis is further analyzed by dividing the sample into two periods viz; pre crisis and post crisis periods. This would provide a better understanding the effect of the global crisis on the Asian countries term structure

Table 8 : Subsample analysis of Level factor before and after the Crisis period

Level _{countires,t} = α +β _{1t} global ^{level} + β _{2t} regional ^{level} +ε _t ^{level}				ε _{countires,t} ^{level} = ε _{countires,t-1} ^{level} +v _t ^{level}		
Precrisis Period :January 2003 – December 2007						
		Global factor	Regional factor	Error	Lag of error term	Residual
Japan	-0.004 (0.007)	0.018* (0.01)	0.04*** (0.005)	ε ^l _{japan,t}	-0.09 (0.14)	0.22 (0.02)
Singapore	4.19 (0.92)	-0.006 (0.03)	0.09*** (0.03)	ε ^l _{Singapore,t}	0.96*** (0.04)	0.17 (0.04)
Hong kong	-0.011 (0.008)	0.035*** (0.006)	0.008*** (0.004)	ε ^l _{Hongkong,t}	0.35*** (0.13)	- 0.07 (0.005)
Korea	-0.00 (0.008)	0.019*** (0.008)	0.016*** (0.005)	ε ^l _{Korea,t}	0.20 (0.13)	0.03 (0.007)
Malaysia	5.03 (0.43)	-0.048 (0.04)	-0.008*** (0.03)	ε ^l _{Malaysia,t}	0.90*** (0.05)	0.05 (0.04)
India	0.001 (0.006)	-0.002 (0.002)	0.002 (0.002)	ε ^l _{india,t}	0.60*** (0.10)	-0.02 (0.002)
China	3.83 (0.57)	-0.013 (0.02)	-0.005 (0.02)	ε ^l _{china,t}	0.95***	0.093 (0.03)
Philippines	- 0.012 0.010)	-0.004 (0.007)	-0.001 (0.005)	ε ^l _{philiphines,t}	0.41 *** (0.02)	0.02 (0.005)
Indonesia	- 0.002 (0.004)	-0.006 (0.008)	0.003 (0.003)	ε ^l _{indonesia,t}	-0.79*** (0.08)	0.96 (0.008)
POST CRISIS PERIOD- January 2008- December 2013						
Japan	-0.005 (0.007)	0.021*** (0.005)	0.014*** (0.005)	ε ^l _{japan,t}	0.16* (0.08)	0.22 (0.007)
Singapore	4.62 (0.52)	0.022 (0.005)	0.065* (0.004)	ε ^l _{Singapore,t}	0.92*** (0.05)	0.09 (0.03)
Hongkong	0.004 (0.018)	0.019*** (0.009)	0.046*** (0.009)	ε ^l _{Hongkong,t}	0.375*** (0.11)	- 0.16 _t (0.11)
Korea	-0.004 (0.004)	0.011*** (0.004)	0.009*** (0.003)	ε ^l _{Korea,t}	-0.12 (0.1)	- 0.07 (0.005)
Malaysia	4.74 (0.25)	-0.011 (0.01)	0.018 (0.02)	ε ^l _{Malaysia,t}	0.89*** (0.05)	0.17 (0.01)
India	-0.0006 (0.003)	0.004** (0.002)	0.0005 (0.002)	ε ^l _{india,t}	0.33*** (0.07)	- 0.03 (0.001)
China	4.12 (0.12)	-0.007 (0.014)	0.02 (0.01)	ε ^l _{china,t}	0.83 (0.06)	0.09 (0.02)

Philippines	- 0.005 (0.013)	0.007 (0.004)	-0.010** (0.005)	$\varepsilon_{philippines,t}^l$	0.56*** (0.10)	-0.04 (0.02)
Indonesia	- 0.001 (0.006)	0.012* (0.007)	0.005 (0.004)	$\varepsilon_{indonesia,t}^l$	-0.46*** (0.10)	0.15 (0.009)

Notes: The table reports the estimates of global and regional influence on country's slope factor. l_{gt} represents the global slope factor, l_{rt} represents the regional slope factor. ε^l represents the error term. v^l is defined as u_{it}/σ_i . Pre crisis period includes January 2003 to December 2007. Post crisis period include January 2008 to December 2013. The figure in the parenthesis represents the standard errors. *, ** and *** shows significance at 10%, 5% and 1 % respectively.

During the pre-crisis period it is found that Indonesia and Philippines is not influenced by the global and regional movements but were found influenced in the crisis affected period. Indonesian bonds are the region's worst hit during the Global financial crisis. The yield rates of Indonesian bonds rose to 16 to 17% in 2008. This result is consistent with the study of Miyajima (2014) who found that during 2008, the reduction in the yield differential among the Indonesian bonds coincided with the US unconventional monetary easing. However the rise in the risk premium of Philippines bonds are balanced by lower rate of expected currency depreciation.

In the post crisis period it is found that Indian long term interest rates are influenced by the global movements. During the Global financial crisis, India witnessed heavy sell off by foreign institutional investors in Equity markets. The reversal of capital inflows and reduced access of Indian entities to international funds has led to reduction in the policy rates which in turn has impact on the long term interest rates.

5.3. Association across short term interest rates

Slope of term structure represents the short term interest rates. It is the policy channel where the monetary authorities change the short term rates to manage domestic growth and inflation. Thus the co movement in the short term rates represents the co movement in the economic fundamentals (Jotikasthira et al.2015). If the global business cycle is synchronized with Asian economies business cycle, it is expected that short term rates would be correlated with the global movements. (Kumar and Okimoto, 2011)

Table 9 : Estimates of Global and regional slope parameters

$\text{Slope}_{countires,t} = \alpha + \beta_{1t} \text{global}^{\text{slope}} + \beta_{2t} \text{regional}^{\text{slope}} + \varepsilon_t^{\text{slope}}$				$\varepsilon_{countires,t}^{\text{slope}} = \varepsilon_{countires,t-1}^{\text{slope}} + v_t^{\text{slope}}$		
Countries	Intercept	Global factor	Regional factor	Error	Lag of error term	Residual

Japan	-2.94 (0.51)	0.028 (0.03)	- 0.027 (0.02)	$\varepsilon_{japan,t}^l$	0.96 *** (0.02)	-0.08 (0.02)
Singapore	-3.74 (1.2)	0.024 (0.05)	-0.013 (0.03)	$\varepsilon_{Singapore,t}^l$	0.97 *** (0.02)	-0.08 (0.03)
Hongkong	-2.51 (0.95)	-0.044 (0.06)	0.029 (0.04)	$\varepsilon_{Hongkong,t}^l$	0.96 *** (0.02)	-0.04 (0.03)
Korea	-1.33 (0.42)	0.046 (0.10)	- 0.033 (0.07)	$\varepsilon_{Korea,t}^l$	0.86*** (0.02)	*** -0.002 (0.04)
Malaysia	-2.12 (0.58)	-0.078* (0.04)	0.052* (0.03)	$\varepsilon_{Malaysia,t}^l$	0.95*** (0.02)	-0.15
India	-0.585 (1.58)	0.015 (0.06)	- 0.021 (0.04)	$\varepsilon_{india,t}^l$	0.97*** (0.02)	0.02 (0.03)
China	-1.46 (0.44)	0.0288 (0.04)	- 0.018 (0.03)	$\varepsilon_{china,t}^l$	0.94*** (0.03)	-0.002 (0.02)
Philippines	-5.00 (0.60)	-0.06 (0.09)	0.045 (0.06)	$\varepsilon_{philippines,t}^l$	0.91*** (0.03)	-0.27 (0.05)
Indonesia	0.026 (0.01)	0.009 (0.02)	- 0.003 (0.04)	$\varepsilon_{indonesia,t}^l$	-0.77*** (0.03)	-0.03 (0.03)

Notes: The table reports the estimates of global and regional influence on country's slope factor. l_{gr} represents the global slope factor, l_r represents the regional slope factor. ε^l represents the error term. v^l is defined as u_{it}/σ_i . The figure in the parenthesis represents the standard errors. *, ** and *** shows significance at 10%, 5% and 1 % respectively.

Table 9 provides the results of the global and regional factor influence on the slope of the yield curve among the Asian economies. There is mix of positive and negative loading upon the global factors. Japan, Korea, Singapore, India , china and Indonesia loads positively upon the global slope factors where as Hong Kong ,Malaysia and Philippines loads negatively upon the global factors. The error terms represents the influence of idiosyncratic factors of domestic bond market. The domestic factors were found statistically significant for all the Asian countries, indicating the influence of domestic monetary policy. The loadings on the global and regional factors are smaller than the country specific factors for all the Asian countries. These results are consistent with the study of Miyajima et al. (2014) who found that US Monetary policy spills over to Asia only in Long term rates and financial integration does not reduce the control of monetary authorities on short term rates. The slope is country specific and sensitive to the business cycle in the economy; it reflects the current stance of the monetary policy. The results indicate despite the spillover effects and financial contagion across the financial market, the domestic monetary authorities were exercising the strong control in the short end of the yield curve.

5.4. Spillovers from US

The short term interest rates which replicate the stance of monetary policy are found to be independent of the global and regional movements. However the Federal Reserve announcement of tapering in 2013 has demonstrated how the changes in bond yields in US are transmitted to the other economies. This necessitates in exploring the reaction of Asian short term rates to the changes in the US policy rates. Among Asian markets, Japan evolved as the dominant economy influencing other markets in the Asian region. Further analysis is attempted here replacing the global and regional factors with US and Japan slope factor respectively.

Table 10. Influence of US and Japan slope factor on Asian Economies

	Full sample (2003 – 2013)		Pre Crisis (2003- 2007)		Post Crisis (2008-2013)	
	US	Japan	US	Japan	US	Japan
Singapore	0.037 (0.07)	-0.76 (1.60)	0.036 (0.07)	-0.73 (1.57)	0.213*** (0.07)	0.547* (0.09)
HongKong	0.003 (0.008)	0.44** (0.20)	0.002 (0.01)	0.31 (0.34)	0.81*** (0.15)	0.15 (0.20)
Korea	-0.274 (0.16)	2.46 (3.87)	-0.26 (0.24)	2.44 (7.47)	-1.44 (2.08)	3.66 (2.90)
Malaysia	0.05*** (0.01)	0.32 (0.23)	0.05 *** (0.01)	0.77** (0.33)	0.439** (0.15)	-0.59* (0.35)
India	0.01 (0.08)	0.97 (2.05)	0.02 (0.08)	0.34 (0.28)	5.5** (2.3)	2.30 (4.07)
China	-0.002 (0.02)	0.36 (0.78)	0.0007 (0.006)	-0.25 (0.39)	-1.56 (0.95)	1.67 (1.3)
Philippines	0.006 (0.005)	0.10 (0.14)	0.006 (0.005)	-0.018 (0.18)	0.099 (0.18)	0.184 (0.25)
Indonesia	0.012 (0.04)	-0.69 (1.14)	0.002 (0.007)	-0.08 (0.22)	2.66* (1.6)	-1.43 (2.2)

Notes: The Table reports regression coefficients of the slope factor regressed with US and Japan Slope factor. The figure in the parenthesis represents the standard errors. *, ** and *** shows significance at 10%, 5% and 1 % respectively.

Table 10 reports influence of US and Japan slope factors on the Asian economies. It was found that Singapore is influenced by US and Japan slope factor in the post crisis period. This indicates Singapore monetary policy is influenced by US in the post crisis period. Hong Kong slope factor is influenced by Japan; however it is influenced by US in the post crisis period. Malaysian slope factor is influenced by the US. Among the emerging economies, the short term rates of India and Indonesia is influenced by US in the post crisis period. Thus the result indicates there is substantial influence of US among the Asian countries in the times of crisis.

5.5. Robustness Check

The robustness of dynamic factor model in extraction of global and regional factor is examined using principal component analysis. The global and regional factors are extracted using principal component analysis. The factors extracted using dynamic factor model and principal components analysis is graphed in chart 1. Both the methods yielded the same results.

Fig 1: Comparison of First component of PCA and the Level factor

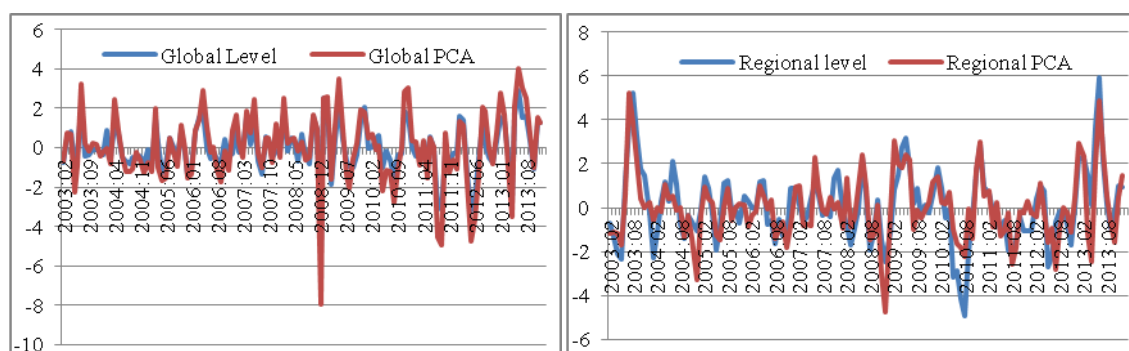
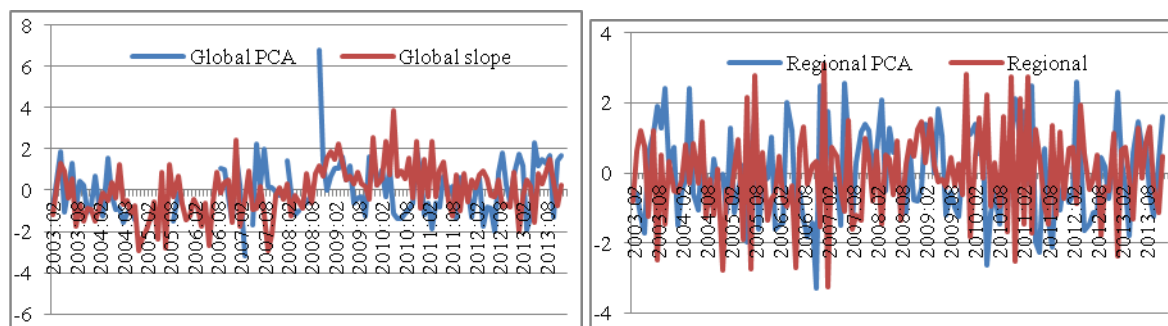


Fig 2: Comparison of First component of PCA and the slope factor



6. Conclusion

The paper investigates the global and regional influence on domestic yieldcurve of Asian countries. The countries term structure of interest rate is modeled by estimating Dynamic Nelson Siegel model as in Diebold et al.2006. The latent factors of the model viz (level, slope and curvature) are extracted using Kalman filter. The global and regional factors are extracted using dynamic factor model. The global factor captures the commonality of US, UK, Germany and

Australia latent factors and regional factor captures the commonality of Japan, Singapore, Hong Kong and Malaysia latent factors. It is found that Japan, South Korea and Hong Kong is influenced by the global and regional factors as these markets are more open and integrated. Singapore and Philippines long term interest rates are influenced by the regional factors. India and China continue to be independent from the global and regional movements. This result provides evidence to the premise that long term interest rates of open economies are more correlated than the short term interest rates. The slope factor of all the Asian countries diverges as they are more independent. This coincides with the premise that slope factor is driven the domestic economic fundamentals and the current stance of monetary policy. The financial integration doesn't reduce the control of monetary authorities on short term interest rates. However it is found that Singapore and Hong Kong short term rates are influenced by US in the post crisis period. The results of the study also found that long term interest rates are the potential transmission channel through which the crisis spreads.

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